

# A Catalan Classification of Extremal Even-Intersecting Lattice Paths

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## Abstract

Shankar proved that a family of North-East lattice paths from  $(0,0)$  to  $(n,n)$  in which every two distinct paths have an even number of common edges has size at most  $2^n$ . He constructed  $C_n$  extremal families from the noncrossing perfect matchings of  $[2n]$  and conjectured that these are all the extremal families. This note proves the conjecture. The main step is an equality classification for the quadratic lemma underlying Shankar's upper bound: its equality cases are indexed by Dyck prefixes. Two auxiliary rigidity statements, one for these Dyck-prefix partitions and one for the parity kernel on the Boolean cube, then determine the extremal relation completely.

**Keywords.** lattice paths, extremal set theory, Eventown, Catalan numbers, noncrossing matchings.

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## 1 Introduction

A North-East lattice path from  $(0,0)$  to  $(n,n)$  is a word in  $\{E, N\}^{2n}$  with  $n$  copies of each letter. Two such paths are *even-intersecting* if the number of common lattice edges is even. Shankar [1] proved that every pairwise even-intersecting family has size at most  $2^n$ . For a noncrossing perfect matching  $M$  of  $[2n]$ , he also considered

$$\mathcal{F}_M = \{w \in \{E, N\}^{2n} : \{w_i, w_j\} = \{E, N\} \text{ for every } \{i, j\} \in M\}. \quad (1)$$

Each  $\mathcal{F}_M$  has size  $2^n$  and is pairwise even-intersecting [1, Section 4]. Since the noncrossing perfect matchings of  $[2n]$  are counted by the Catalan number  $C_n$  (see, for example, [2]), this gives at least  $C_n$  extremal families. Shankar conjectured that there are no others [1, Conjecture 4.1] and verified the conjecture for  $n \leq 5$  [1, Section 4].

The purpose of this note is to prove the conjecture in its stronger structural form.

**Theorem 1.1.** *Let  $n \geq 1$ , and let  $\mathcal{F}$  be a family of lattice paths from  $(0,0)$  to  $(n,n)$  such that every two distinct members have an even number of common edges. If  $|\mathcal{F}| = 2^n$ , then there is a unique noncrossing perfect matching  $M$  of  $[2n]$  for which*

$$\mathcal{F} = \mathcal{F}_M.$$

*Consequently there are exactly  $C_n$  extremal families.*

The proof follows the equality structure in Shankar's argument rather than introducing a different upper bound. It has three ingredients. First, equality in the quadratic lemma from [1] is classified by Dyck prefixes. Second, two such equality partitions cannot be cross-compatible unless their Dyck prefixes have the same matched edges. Third, a weight-preserving automorphism of the Boolean cube that preserves the path-intersection parity kernel is trivial. The essential point is to retain the full equality information throughout the induction.

## 2 The parity kernel and Dyck-prefix partitions

Identify an  $m$ -step path with a word  $x = x_1 \cdots x_m \in \{0, 1\}^m$ , where  $1 = N$  and  $0 = E$ . Write  $\text{wt}(x) = \sum_i x_i$ . Following [1], put

$$B_m(x, y) = |E(x) \cap E(y)| \pmod{2} \in \mathbb{F}_2.$$

Equivalently,

$$B_m(x, y) = \sum_{i=1}^m \mathbf{1}_{\{x_i=y_i\}} \mathbf{1}_{\{\sum_{j<i} x_j = \sum_{j<i} y_j\}} \quad \text{in } \mathbb{F}_2. \quad (2)$$

In particular  $B_m(x, x) = m \pmod{2}$ .

The last coordinate gives the basic recurrence. If  $p, q \in \{0, 1\}^{m-1}$  and  $\alpha, \beta \in \{0, 1\}$ , then

$$B_m(p\alpha, q\beta) = B_{m-1}(p, q) + \mathbf{1}_{\{\alpha=\beta\}} \mathbf{1}_{\{\text{wt}(p)=\text{wt}(q)\}}. \quad (3)$$

### 2.1 Dyck-prefix blocks

A *Dyck prefix* of length  $m$  is a word  $P = P_1 \cdots P_m$  in the alphabet  $\{U, D\}$  whose height never becomes negative. Match every  $D$  to the most recent unmatched  $U$ . Let  $\mathcal{E}(P)$  be the resulting noncrossing partial matching on  $[m]$ , and let

$$u_1 < \cdots < u_r$$

be the unmatched positions. If  $k = |\mathcal{E}(P)|$ , then  $m = 2k + r$ .

For  $\varepsilon = (\varepsilon_1, \dots, \varepsilon_r) \in \{0, 1\}^r$  define

$$\mathcal{A}_\varepsilon(P) = \{x \in \{0, 1\}^m : x_a \neq x_b \text{ for every } \{a, b\} \in \mathcal{E}(P), \quad x_{u_i} = \varepsilon_i \ (1 \leq i \leq r)\}. \quad (4)$$

Thus every block has size  $2^k$  and lies in the weight level  $k + \text{wt}(\varepsilon)$ . Put

$$\Omega(P) = \bigcup_{\varepsilon \in \{0, 1\}^r} \mathcal{A}_\varepsilon(P).$$

The next cancellation fact is the reason these blocks appear naturally.

**Lemma 2.1** (cancellation). *Let  $P$  be a Dyck prefix of length  $m$  with  $r$  unmatched positions. Suppose  $x, y \in \Omega(P)$ , and delete from both words every coordinate belonging to an edge of  $\mathcal{E}(P)$ . If the resulting words are  $\bar{x}, \bar{y} \in \{0, 1\}^r$ , then*

$$B_m(x, y) = B_r(\bar{x}, \bar{y}). \quad (5)$$

*In particular, if  $x \in \mathcal{A}_\varepsilon(P)$  and  $y \in \mathcal{A}_\eta(P)$ , then*

$$B_m(x, y) = B_r(\varepsilon, \eta). \quad (6)$$

*Proof.* No unmatched position of a Dyck prefix can lie strictly inside a matched edge: otherwise it would still be on top of the stack when that edge closed. Hence every nonempty matching  $\mathcal{E}(P)$  contains an innermost edge  $\{i, i+1\}$ . Both  $x$  and  $y$  use one 0 and one 1 at these two positions. If the orientations agree, the two positions contribute either two common edges or none; if the orientations disagree, they contribute none. Hence their contribution is even. After the two positions are deleted, both paths have been translated by the same vector  $(1, 1)$  from that point onward, so all later common-edge incidences are unchanged. Deleting innermost edges successively proves (5).  $\square$

It follows at once that the blocks (4) satisfy the three compatibility conditions that will occur in Section 3: within a block the value of  $B_m$  is  $r \equiv m \pmod{2}$ , while on a product of two blocks it is constant.

We shall also need the following two-step test.

**Lemma 2.2** (square test). *Let  $s \geq 1$  and  $y = y'\beta\gamma \in \{0,1\}^{s+1}$  with  $y' \in \{0,1\}^{s-1}$ . The following are equivalent:*

- (i)  $B_{s+1}(z10, y) = B_{s+1}(z01, y)$  for every  $z \in \{0,1\}^{s-1}$ ;
- (ii)  $\beta \neq \gamma$ .

*Proof.* Let  $h = \text{wt}(z)$  and  $q = \text{wt}(y')$ . Applying (3) twice gives

$$\begin{aligned} B_{s+1}(z10, y'\beta\gamma) + B_{s+1}(z01, y'\beta\gamma) \\ = \mathbf{1}_{\{h=q\}} + \mathbf{1}_{\{\gamma=0\}}\mathbf{1}_{\{h=q+\beta-1\}} + \mathbf{1}_{\{\gamma=1\}}\mathbf{1}_{\{h=q+\beta\}}. \end{aligned}$$

If  $\beta \neq \gamma$ , the right-hand side is identically zero. If  $\beta = \gamma = 0$ , take  $h = q$ ; if  $\beta = \gamma = 1$ , again take  $h = q$ . Since  $0 \leq q \leq s-1$ , such a word  $z$  exists, and the displayed value is then 1.  $\square$

## 2.2 Cross-rigidity

For Dyck prefixes  $P, Q$  of the same length, write  $P \preccurlyeq Q$  if, for every block  $A$  of the partition  $\{\mathcal{A}_\varepsilon(P)\}$ , every  $x, x' \in A$ , and every  $y \in \Omega(Q)$ ,

$$B_m(x, y) = B_m(x', y). \quad (7)$$

Thus changing the orientations of the matched edges of  $P$  is invisible to every word satisfying the matching constraints of  $Q$ .

**Lemma 2.3.** *For Dyck prefixes  $P, Q$  of length  $m$ ,*

$$P \preccurlyeq Q \implies \mathcal{E}(P) \subseteq \mathcal{E}(Q).$$

*Proof.* Proceed by induction on  $m$ . The case  $m = 0$  is immediate. Assume  $m \geq 1$ , and remove the last symbol from  $P$  and  $Q$ , obtaining Dyck prefixes  $P^-$  and  $Q^-$ . Every word in  $\Omega(Q^-)$  can be extended to a word in  $\Omega(Q)$ : if the last symbol of  $Q$  is  $U$ , choose either last bit; if it is  $D$ , the last bit is forced to be opposite to the bit at its matched opener. Likewise, any two words in the same block for  $P^-$  can be extended with a common last bit to two words in one block for  $P$ ; when the last symbol of  $P$  is  $D$ , the bit at its opener is fixed inside a block of  $P^-$ . Since all words in one block have the same weight, the correction term in (3) is the same for the two extensions. Hence

$$P^- \preccurlyeq Q^-.$$

By induction,  $\mathcal{E}(P^-) \subseteq \mathcal{E}(Q^-)$ .

If the last symbol of  $P$  is  $U$ , this already gives  $\mathcal{E}(P) \subseteq \mathcal{E}(Q)$ . Suppose instead that it is  $D$ , and let  $e$  be its matched edge. Delete the endpoints of all edges in  $\mathcal{E}(P^-)$ . By induction these edges also belong to  $\mathcal{E}(Q^-)$ . Lemma 2.1, applied successively to the same matched pairs on both sides of (7), shows that every defining  $B$ -equality is unchanged; hence  $\preccurlyeq$  descends to the reduced prefixes. Removing the endpoints of a matched  $U$ – $D$  pair preserves the Dyck-prefix property. On the  $P$ -side this is equivalently the cancellation associated with  $P^-U$ , whose matching is exactly  $\mathcal{E}(P^-)$ . The reduced prefix coming from  $P$  has the form

$$U^s D$$

for some  $s \geq 1$ : all surviving vertices before the final  $D$  are unmatched, and the last of them is its opener. Let  $Q'$  be the reduced prefix coming from  $Q$ .

A block of  $U^s D$  is  $\{z10, z01\}$  for some  $z \in \{0, 1\}^{s-1}$ . Thus, for every  $y \in \Omega(Q')$ ,

$$B_{s+1}(z10, y) = B_{s+1}(z01, y) \quad \text{for all } z \in \{0, 1\}^{s-1}.$$

Lemma 2.2 gives  $y_s \neq y_{s+1}$  for every  $y \in \Omega(Q')$ . In a Dyck-prefix matching, two positions have opposite bits in every word of  $\Omega(Q')$  if and only if they are matched to one another: if they lie on different matching edges, or if one of them is unmatched, the relevant orientations and unmatched bits can be chosen independently to make the two bits equal. Therefore  $\{s, s+1\} \in \mathcal{E}(Q')$ . Lifting this edge back gives  $e \in \mathcal{E}(Q)$ .  $\square$

**Proposition 2.4** (cross-rigidity). *Let  $P, Q$  be Dyck prefixes of length  $m$ . Suppose that  $B_m$  is constant on the product of every block of the  $P$ -partition with every block of the  $Q$ -partition. Then  $P = Q$ .*

*Proof.* The hypothesis gives  $P \preceq Q$ . Since  $B_m$  is symmetric, it also gives  $Q \preceq P$ . Lemma 2.3 yields  $\mathcal{E}(P) = \mathcal{E}(Q)$ . A Dyck prefix is determined by its matching edges: the right endpoints are the  $D$ -positions, and every other position is a  $U$ -position. Hence  $P = Q$ .  $\square$

### 3 Equality in the quadratic lemma

Shankar's proof of the even-intersection bound rests on the following inequality [1, Lemma 2.3]. Empty members do not affect the inequality and may be discarded. The equality classification is stated simultaneously.

**Theorem 3.1** (quadratic equality classification). *Let  $A_1, \dots, A_t$  be nonempty pairwise disjoint subsets of  $\{0, 1\}^m$  such that*

- (i) *each  $A_i$  is contained in one Hamming-weight level;*
- (ii)  *$B_m(x, y) = m \pmod{2}$  for all  $x, y \in A_i$ ;*
- (iii) *for  $i \neq j$ , the function  $B_m$  is constant on  $A_i \times A_j$ .*

*Then*

$$\sum_{i=1}^t |A_i|^2 \leq 2^m. \tag{8}$$

*Equality holds if and only if there is a Dyck prefix  $P$  of length  $m$  such that, up to reordering,*

$$\{A_1, \dots, A_t\} = \{\mathcal{A}_\varepsilon(P) : \varepsilon \in \{0, 1\}^r\},$$

*where  $r$  is the number of unmatched positions of  $P$ . The prefix  $P$  is unique.*

The equality analysis reduces to the following local Boolean-cube classification. The same lemma will later control automorphisms.

For  $x, y \in \{0, 1\}^d$ , set

$$\Delta_{x,y}(z) = B_d(x, z) + B_d(y, z) \in \mathbb{F}_2.$$

Let  $\mathcal{L}_j^{(d)} = \{z \in \{0, 1\}^d : \text{wt}(z) = j\}$ , with the convention that this set is empty if  $j \notin \{0, \dots, d\}$ .

**Lemma 3.2** (localized pairs). *Let  $u, v \in \{0, 1\}^d$  satisfy  $\text{wt}(u) = k + 1$  and  $\text{wt}(v) = k$ . Assume*

$$\Delta_{u,v}(z) = 0 \quad \text{whenever } \text{wt}(z) \notin \{k, k + 1\}. \quad (9)$$

*Then either*

$$u = t1, \quad v = t0 \quad (10)$$

*for some  $t \in \{0, 1\}^{d-1}$ , or  $d = 2k + 1 \geq 3$  and*

$$u = 1(10)^k, \quad v = 0(01)^k. \quad (11)$$

*For the exceptional pair,*

$$\Delta_{u,v}(z) = 1 \iff \text{wt}(z) \in \{k, k + 1\}. \quad (12)$$

*Proof.* Write  $u = p\alpha$ ,  $v = q\beta$ , and  $z\gamma$  for the test word. From (3),

$$\Delta_{p\alpha, q\beta}(z\gamma) = \Delta_{p,q}(z) + \mathbf{1}_{\{\alpha=\gamma\}} \mathbf{1}_{\{\text{wt}(p)=\text{wt}(z)\}} + \mathbf{1}_{\{\beta=\gamma\}} \mathbf{1}_{\{\text{wt}(q)=\text{wt}(z)\}}. \quad (13)$$

The proof is a finite-state induction; the details below account for the exceptional alternating state.

First consider the following auxiliary states.

- $S_d(k)$ :  $\text{wt}(x) = \text{wt}(y) = k$  and  $\text{supp } \Delta_{x,y} \subseteq \mathcal{L}_k^{(d)}$ ;
- $E_d(k)$ :  $\text{wt}(x) = \text{wt}(y) = k$  and  $\Delta_{x,y} = \mathbf{1}_{\mathcal{L}_k^{(d)}}$ ;
- $L_d^-(k)$ :  $\text{wt}(x) = k + 1$ ,  $\text{wt}(y) = k$ , and  $\text{supp } \Delta_{x,y} \subseteq \mathcal{L}_k^{(d)}$ ;
- $L_d^+(k)$ : the same weight condition, with support contained in  $\mathcal{L}_{k+1}^{(d)}$ .

The case  $d = 0$  is immediate. A simultaneous induction on  $d$ , using (13), gives

$$S_d(k) \implies x = y, \quad E_d(k), L_d^-(k), L_d^+(k) \text{ are impossible.} \quad (14)$$

The transition rules are as follows. A terminal pattern whose implied prefix weight lies outside  $\{0, \dots, d - 1\}$  is infeasible; whenever a pattern is feasible, each weight level used in the argument below is nonempty. In the state  $E_d(k)$ , equal terminal bits are impossible. For terminal pattern 00, take a prefix word  $z$  of weight  $k$ : the two choices  $z0, z1$  have target values 1, 0, whereas the left side of (13) is independent of  $\gamma$ . For terminal pattern 11, the same contradiction is obtained from a prefix word of weight  $k - 1$ . Unequal terminal bits force  $\Delta_{p,q} \equiv 0$  for a pair of adjacent weights, contradicting the induction hypothesis for  $L^-/L^+$ . For  $L_d^-(k)$ , the terminal patterns 00 and 01 give an immediate contradiction by evaluating (13) on a word of weight  $k + 1$ ; the pattern 10 reduces to  $E_{d-1}(k)$ , while 11 reduces to  $L_{d-1}^+(k - 1)$ . If  $\bar{x}$  denotes the bitwise complement of  $x$ , then  $B_d(\bar{x}, \bar{z}) = B_d(x, z)$ . Hence simultaneous complementation of the two words and the test word, followed by swapping the first two words, sends an  $L^+$  state to an  $L^-$  state. Finally, in  $S_d(k)$ , equal terminal bits force  $\Delta_{p,q} \equiv 0$ , so induction gives  $p = q$ ; unequal terminal bits again force an adjacent-weight pair with zero profile, contradicting  $L^-/L^+$ .

It remains to track two alternating states. Let

$$\begin{aligned} M_d(k) : & \quad \text{wt}(x) = k + 1, \text{ wt}(y) = k - 1, \quad \text{supp } \Delta_{x,y} \subseteq \mathcal{L}_k^{(d)}, \\ T_d(k) : & \quad \text{wt}(x) = k + 1, \text{ wt}(y) = k, \quad \Delta_{x,y} = \mathbf{1}_{\mathcal{L}_k^{(d)} \cup \mathcal{L}_{k+1}^{(d)}}. \end{aligned}$$

The remaining two states have the following transitions, again by (13). In an  $M_d(k)$  state, the patterns 00 and 01 are contradicted by testing a word of weight  $k+1$ , while 11 is contradicted by testing a word of weight  $k-2$  (the pattern is feasible only when that layer is nonempty). The only surviving pattern is 10, and the prefix then has value 1 on exactly the two layers  $k-1, k$ ; hence it is a  $T_{d-1}(k-1)$  pair. In a  $T_d(k)$  state with  $d > 1$ , the patterns 00 and 11 are contradicted by testing a prefix word of weight  $k$ . For the pattern 10, the two last-bit choices give the same value. If  $k > 0$ , a prefix word of weight  $k-1$  gives target values 0, 1; if  $k = 0$ , a prefix word of weight 1 gives target values 1, 0 (and this level is nonempty because  $d > 1$ ). Thus only 01 survives, and (13) says that its prefix has support exactly on layer  $k$ , so it is an  $M_{d-1}(k)$  pair. Since  $T_1(0)$  consists only of (1, 0), induction gives

$$M_{2k}(k) : (x, y) = (11(01)^{k-1}, 00(10)^{k-1}) \quad (k \geq 1), \quad (15)$$

$$T_{2k+1}(k) : (x, y) = (1(10)^k, 0(01)^k). \quad (16)$$

No  $M$  or  $T$  state occurs in the opposite parity.

Now apply (13) to a pair satisfying (9). According to the terminal pattern  $(\alpha, \beta)$ , its prefix lies in the following state:

| $(\alpha, \beta)$ | prefix state | consequence                |
|-------------------|--------------|----------------------------|
| 00                | $L^-$        | impossible                 |
| 11                | $L^+$        | impossible                 |
| 10                | $S$          | $p = q$                    |
| 01                | $M$          | the alternating pair (11). |

The 10 case is exactly (10). The 01 case, together with (15), yields (11); its profile is (12) by (16). When  $d = 1$ , this alternating pair is the canonical pair itself, so the exceptional alternative is new only for  $d \geq 3$ .  $\square$

We next isolate the reconstruction step that occurs after the two inductive halves have become the same Dyck-prefix partition.

For a family  $\mathcal{C} = \{C_i\}$  of nonempty subsets of  $\{0, 1\}^{r+1}$ , write

$$C_i^a = \{z \in \{0, 1\}^r : za \in C_i\}, \quad a \in \{0, 1\},$$

let  $I(\mathcal{C})$  be the indices for which both  $C_i^0$  and  $C_i^1$  are nonempty, and define

$$\mathcal{G}_0(\mathcal{C}) = \{C_i^0 : C_i^0 \neq \emptyset\} \cup \{C_i^1 : i \in I(\mathcal{C})\},$$

$$\mathcal{G}_1(\mathcal{C}) = \{C_i^1 : C_i^1 \neq \emptyset\} \cup \{C_i^0 : i \in I(\mathcal{C})\}.$$

These are the two split families used in Shankar's induction.

**Lemma 3.3** (reconstruction). *Let  $r \geq 0$ . Suppose a family  $\mathcal{C}$  of nonempty subsets of  $\{0, 1\}^{r+1}$  satisfies the three conditions of Theorem 3.1, and suppose*

$$\mathcal{G}_0(\mathcal{C}) = \mathcal{G}_1(\mathcal{C}) = \{\{z\} : z \in \{0, 1\}^r\}.$$

*Then exactly one of the following holds:*

$$(a) \ \mathcal{C} = \{\{z0\}, \{z1\} : z \in \{0, 1\}^r\};$$

$$(b) \ r \geq 1 \text{ and}$$

$$\mathcal{C} = \{\{t10, t01\} : t \in \{0, 1\}^{r-1}\}.$$

*Proof.* Every nonempty half  $C_i^a$  is a singleton, since it occurs in one of the two displayed singleton partitions. There is also no overlap between opposite halves when one of the corresponding blocks is split. Indeed, suppose  $i \in I(\mathcal{C})$ ,  $j \neq i$ , and

$$z \in C_j^0 \cap C_i^1.$$

Choose  $x \in C_i^0$ . Then  $z1, x0 \in C_i$ , so condition (ii) gives  $B_{r+1}(z1, x0) = r + 1 \pmod{2}$ . Weight homogeneity gives  $\text{wt}(x) = \text{wt}(z) + 1$ , and (3) therefore yields

$$B_{r+1}(z0, x0) = B_{r+1}(z1, x0) = r + 1 \pmod{2}.$$

On the other hand,  $B_{r+1}(z0, z1) = B_r(z, z) = r \pmod{2}$ , contradicting constancy on  $C_j \times C_i$ . The case  $C_j^1 \cap C_i^0 \neq \emptyset$  is symmetric.

It follows that each label  $z \in \{0, 1\}^r$  either contributes the two singleton blocks  $\{z0\}, \{z1\}$ , or occurs in exactly one split block. Such a block has the form

$$\{u0, v1\}, \quad \text{wt}(u) = \text{wt}(v) + 1, \quad (17)$$

where the weight relation follows from condition (i). The labels used in distinct two-element blocks are disjoint.

Fix a block (17), and write  $k = \text{wt}(v)$ . If a label  $z$  is unpaired, cross-constancy with the two singleton blocks  $\{z0\}$  and  $\{z1\}$  gives

$$\Delta_{u,v}(z) = \mathbf{1}_{\{\text{wt}(z)=k+1\}} = \mathbf{1}_{\{\text{wt}(z)=k\}}. \quad (18)$$

If another two-element block is  $\{a0, b1\}$ , comparison of the two rows in its two columns gives

$$\Delta_{u,v}(a) = \mathbf{1}_{\{\text{wt}(a)=k+1\}}, \quad \Delta_{u,v}(b) = \mathbf{1}_{\{\text{wt}(b)=k\}}. \quad (19)$$

Equations (18)–(19) imply that  $\Delta_{u,v}$  vanishes outside the two levels  $k, k + 1$ . Lemma 3.2 therefore says that every paired block is canonical,

$$\{t10, t01\},$$

except possibly for the unique alternating pair in odd dimension.

First exclude that exception. Suppose  $r = 2s + 1 \geq 3$  and

$$u = 1(10)^s, \quad v = 0(01)^s.$$

By (12) and (18)–(19), every word of weight  $s + 1$  must occur as a high endpoint and every word of weight  $s$  as a low endpoint. All other pairs are canonical. A canonical high endpoint ends in 1, while a canonical low endpoint ends in 0. If  $s \geq 2$ , there are  $\binom{2s}{s-1} > 1$  weight- $s$  words ending in 1, so one of them is different from  $v$ ; it can be neither the exceptional low endpoint nor a canonical low endpoint, a contradiction. If  $s = 1$ , the exceptional block is  $\{1100, 0011\}$  and one forced canonical block is  $\{0110, 0101\}$ . Their  $B_4$ -matrix is

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix},$$

contrary to cross-constancy. Thus no exceptional pair occurs.

It remains to show that the canonical pairs are either absent or all present. Suppose the pair indexed by  $t \in \{0, 1\}^{r-1}$  is present, and put  $h = \text{wt}(t)$ . For  $u = t1$  and  $v = t0$ , recurrence (3) gives

$$\Delta_{u,v}(z) = 1 \iff \text{wt}(z_1 \cdots z_{r-1}) = h. \quad (20)$$

No label of total weight  $h$  or  $h + 1$  can be unpaired by (18). Combining (20) with (19) shows that the labels

$$s0 \text{ (wt}(s) = h), \quad s1 \text{ (wt}(s) = h - 1), \quad s1 \text{ (wt}(s) = h), \quad s0 \text{ (wt}(s) = h + 1)$$

must respectively be low, high, high, and low endpoints. Hence every canonical pair whose prefix weight lies in  $\{h - 1, h, h + 1\} \cap \{0, \dots, r - 1\}$  is present. Iterating along the connected sequence of weight levels  $0, 1, \dots, r - 1$  forces all canonical pairs. This proves the dichotomy.  $\square$

*Proof of Theorem 3.1.* Proceed by induction on  $m$ , proving the inequality and its equality statement simultaneously. The case  $m = 0$  is immediate. For every  $i$ , write

$$A_i^a = \{x \in \{0, 1\}^{m-1} : xa \in A_i\}, \quad a \in \{0, 1\},$$

and let  $I$  be the set of indices for which both  $A_i^0$  and  $A_i^1$  are nonempty. Consider the two collections of nonempty sets

$$\begin{aligned} \mathcal{G}_0 &= \{A_i^0 : A_i^0 \neq \emptyset\} \cup \{A_i^1 : i \in I\}, \\ \mathcal{G}_1 &= \{A_i^1 : A_i^1 \neq \emptyset\} \cup \{A_i^0 : i \in I\}. \end{aligned}$$

Each collection satisfies the three hypotheses in dimension  $m - 1$ . The weight condition is immediate, and (3) transfers internal and cross-constancy. It remains to verify disjointness. Suppose, for example, that

$$u \in A_j^0 \cap A_i^1$$

with  $i \in I$ . If  $i = j$ , then  $u0, u1 \in A_i$  but  $B_m(u0, u1) = m - 1$ , contradicting (ii). Thus  $i \neq j$ . Choose  $x \in A_i^0$ . Then  $u1, x0 \in A_i$ , so  $B_m(u1, x0) = m$ . Since these two words have the same total weight,  $\text{wt}(x) = \text{wt}(u) + 1$ , and (3) gives

$$B_m(u0, x0) = B_m(u1, x0) = m.$$

On the other hand  $B_m(u0, u1) = m - 1$ . This contradicts constancy on  $A_j \times A_i$ . The other possible overlap is symmetric.

Put  $a_i = |A_i^0|$  and  $b_i = |A_i^1|$ . By induction,

$$\sum_{i \notin I} a_i^2 + \sum_{i \in I} (a_i^2 + b_i^2) \leq 2^{m-1}, \quad (21)$$

$$\sum_{i \notin I} b_i^2 + \sum_{i \in I} (a_i^2 + b_i^2) \leq 2^{m-1}. \quad (22)$$

For  $i \in I$ ,

$$(a_i + b_i)^2 \leq 2a_i^2 + 2b_i^2,$$

with equality exactly when  $a_i = b_i$ . Adding (21) and (22) proves (8).

Assume now that equality holds. Then both (21) and (22) are equalities and  $a_i = b_i$  for every  $i \in I$ . By induction there are Dyck prefixes  $P, Q$  of length  $m - 1$  such that  $\mathcal{G}_0$  and  $\mathcal{G}_1$  are their block partitions. Moreover  $B_{m-1}$  is constant on every block of  $\mathcal{G}_0$  times every block of  $\mathcal{G}_1$ : if the two prefix blocks come from  $A_i^\alpha$  and  $A_j^\beta$ , this follows from (3), since its correction term is constant on that product and  $B_m$  is constant on  $A_i \times A_j$  (also when  $i = j$ ). Proposition 2.4 yields  $P = Q$ .

Delete from every prefix word the coordinates belonging to  $\mathcal{E}(P)$ . Equivalently, apply Lemma 2.1 to the Dyck prefix  $PU$ , whose last position is unmatched; this preserves all  $B$ -conditions, including the final bit of the original words. Since every block of  $P$  collapses to one label in  $\{0, 1\}^r$ , the reduced reconstruction is exactly the situation of Lemma 3.3. In case

(a) the new Dyck prefix is  $PU$ ; in case (b) it is  $PD$ . This completes the induction and proves existence. Conversely, if  $P$  has  $k$  matched edges and  $r$  unmatched positions, Lemma 2.1 shows that its blocks satisfy (i)–(iii), and

$$\sum_{\varepsilon \in \{0,1\}^r} |\mathcal{A}_\varepsilon(P)|^2 = 2^r (2^k)^2 = 2^{2k+r} = 2^m.$$

Thus every Dyck-prefix partition is indeed an equality case.

For uniqueness, note that from the union  $\Omega(P)$  one recovers the matching by

$$\{i, j\} \in \mathcal{E}(P) \iff x_i \neq x_j \text{ for every } x \in \Omega(P).$$

If  $i, j$  are not matched, their bits can be made equal by independently orienting their matching edges and choosing unmatched bits. Hence the equality partition determines  $P$ .  $\square$

## 4 Rigidity on the Boolean cube

The final equality step produces a bijection of the unmatched labels. The next proposition shows that this bijection cannot be nontrivial.

**Proposition 4.1** (cube rigidity). *Let  $\phi : \{0, 1\}^r \rightarrow \{0, 1\}^r$  be a bijection satisfying*

$$\text{wt}(\phi(x)) = \text{wt}(x), \quad B_r(\phi(x), \phi(y)) = B_r(x, y) \quad (23)$$

*for all  $x, y \in \{0, 1\}^r$ . Then  $\phi$  is the identity.*

*Proof.* For words of adjacent weights, define an intrinsic relation

$$u \sim v \iff \text{wt}(u) = \text{wt}(v) + 1 \text{ and } \Delta_{u,v}(z) = 0 \text{ outside the two levels } \text{wt}(v), \text{wt}(u).$$

Lemma 3.2 identifies all edges of this relation. They are the canonical edges

$$t1 \sim t0 \quad (t \in \{0, 1\}^{r-1}), \quad (24)$$

plus, when  $r \geq 3$  is odd, the single exceptional edge

$$1(10)^{(r-1)/2} \sim 0(01)^{(r-1)/2}.$$

The relation is preserved by  $\phi$ . Indeed,

$$\Delta_{\phi(u), \phi(v)}(\phi(z)) = \Delta_{u,v}(z)$$

for every  $z$ , and  $\phi$  preserves Hamming-weight levels.

If  $r$  is even, (24) is therefore preserved as a set. If  $r \geq 3$  is odd, the two endpoints of the exceptional edge are the only vertices of degree two in the relation graph; hence the exceptional edge is intrinsically distinguished, and after removing it the canonical matching is again preserved. The cases  $r = 0, 1$  are immediate. Thus in all cases  $\phi$  maps every canonical pair to a canonical pair. Weight preservation prevents the two endpoints from being exchanged, so there is a weight-preserving bijection  $\psi : \{0, 1\}^{r-1} \rightarrow \{0, 1\}^{r-1}$  such that

$$\phi(t\alpha) = \psi(t)\alpha \quad (t \in \{0, 1\}^{r-1}, \alpha \in \{0, 1\}).$$

Since the last bits differ,

$$B_r(t0, s1) = B_{r-1}(t, s).$$

Using (23),

$$B_{r-1}(\psi(t), \psi(s)) = B_r(\phi(t0), \phi(s1)) = B_r(t0, s1) = B_{r-1}(t, s).$$

Hence  $\psi$  satisfies the same hypotheses in dimension  $r - 1$ . Induction gives  $\psi = \text{id}$  and therefore  $\phi = \text{id}$ .  $\square$

## 5 Extremal relations

Call a relation  $R \subseteq \{0, 1\}^n \times \{0, 1\}^n$  *admissible* if

$$\text{wt}(x) = \text{wt}(y) \quad ((x, y) \in R) \quad (25)$$

and

$$B_n(x, x') = B_n(y, y') \quad \text{for all } (x, y), (x', y') \in R. \quad (26)$$

Shankar proved that every admissible relation has at most  $2^n$  elements [1, Lemma 2.2]. Theorem 3.1 gives its complete equality case.

**Proposition 5.1.** *If  $R \subseteq \{0, 1\}^n \times \{0, 1\}^n$  is admissible and  $|R| = 2^n$ , then there are Dyck prefixes  $P, Q$  of length  $n$ , with the same number  $r$  of unmatched positions, such that*

$$R = \bigcup_{\varepsilon \in \{0, 1\}^r} \mathcal{A}_\varepsilon(P) \times \mathcal{A}_\varepsilon(Q). \quad (27)$$

*Proof.* Form the bipartite graph whose edges are the pairs in  $R$ , and ignore isolated vertices. Let  $A_j$  and  $B_j$  be the left and right vertex sets of its connected components. If two left vertices  $x, x'$  have a common neighbor  $y$ , then (26) gives

$$B_n(x, x') = B_n(y, y) = n \pmod{2}.$$

An induction on graph distance along an alternating path shows the same for any two vertices in one  $A_j$ , and similarly in one  $B_j$ : if  $x_0 - y_1 - x_1 - \dots - y_\ell - x_\ell$  is a shortest alternating path, then (26) gives  $B_n(x_0, x_\ell) = B_n(y_1, y_\ell)$ , and the latter pair has smaller distance. Also, (25) makes each  $A_j$  and  $B_j$  weight-homogeneous. Finally, if  $A_i$  and  $A_j$  are different components, then  $B_n$  is constant on  $A_i \times A_j$ . Indeed, if  $x, x' \in A_i$  share a neighbor  $y$  and  $(a, b)$  is any edge of the other component, then

$$B_n(x, a) = B_n(y, b) = B_n(x', a).$$

Connectivity lets one move either endpoint, so the value is constant. The same holds on the right. Thus both collections satisfy Theorem 3.1.

The standard component estimate is

$$|R| \leq \sum_j |A_j| |B_j| \leq \left( \sum_j |A_j|^2 \right)^{1/2} \left( \sum_j |B_j|^2 \right)^{1/2} \leq 2^n. \quad (28)$$

Since  $|R| = 2^n$ , equality holds throughout. In particular every component is complete bipartite, and both quadratic sums equal  $2^n$ . Theorem 3.1 gives Dyck prefixes  $P, Q$  whose block partitions are  $\{A_j\}$  and  $\{B_j\}$ . If their numbers of unmatched positions are  $r_P, r_Q$ , then the common number of components is both  $2^{r_P}$  and  $2^{r_Q}$ , so  $r_P = r_Q =: r$ . Consequently  $P$  and  $Q$  also have the same number  $k = (n - r)/2$  of matched edges.

Index the blocks of  $P$  by  $\varepsilon \in \{0, 1\}^r$ . The components give a bijection  $\phi : \{0, 1\}^r \rightarrow \{0, 1\}^r$  such that

$$R = \bigcup_{\varepsilon \in \{0, 1\}^r} \mathcal{A}_\varepsilon(P) \times \mathcal{A}_{\phi(\varepsilon)}(Q).$$

A word in  $\mathcal{A}_\varepsilon(P)$  has weight  $k + \text{wt}(\varepsilon)$ , and a word in  $\mathcal{A}_{\phi(\varepsilon)}(Q)$  has weight  $k + \text{wt}(\phi(\varepsilon))$ . Completeness of the component and (25) therefore give

$$\text{wt}(\phi(\varepsilon)) = \text{wt}(\varepsilon).$$

For labels  $\varepsilon, \eta$ , choose arbitrary words in the two corresponding complete components. Equation (26), followed by Lemma 2.1, gives

$$B_r(\phi(\varepsilon), \phi(\eta)) = B_r(\varepsilon, \eta).$$

Proposition 4.1 gives  $\phi = \text{id}$ , proving (27).  $\square$

## 6 Proof of the Catalan classification

Let  $F = f_1 \cdots f_{2n}$  be a path from  $(0, 0)$  to  $(n, n)$ , viewed as a balanced binary word, and define

$$u(F) = (f_1, \dots, f_n), \quad v(F) = (1 - f_{2n}, \dots, 1 - f_{n+1}). \quad (29)$$

Then  $\text{wt}(u(F)) = \text{wt}(v(F))$ . Conversely every pair  $(u, v)$  of equal weight reconstructs a unique balanced word by taking first half  $u$  and second half the reversed complement of  $v$ . Thus (29) is a bijection between paths from  $(0, 0)$  to  $(n, n)$  and equal-weight pairs in  $\{0, 1\}^n \times \{0, 1\}^n$ .

Reversing and complementing the second half is a reflection of that half-path and preserves common edges. Therefore, for any two balanced paths  $F, G$ ,

$$B_{2n}(F, G) = B_n(u(F), u(G)) + B_n(v(F), v(G)) \quad \text{in } \mathbb{F}_2, \quad (30)$$

which is Observation 2.1 of [1].

*Proof of Theorem 1.1.* Let  $\mathcal{F}$  be extremal and let

$$R = \{(u(F), v(F)) : F \in \mathcal{F}\}.$$

By (30), the even-intersection hypothesis says exactly that  $R$  is admissible for distinct pairs; for a pair with itself, both sides of (26) equal  $n \pmod{2}$ , so admissibility holds in full. Since  $|R| = 2^n$ , Proposition 5.1 gives Dyck prefixes  $P, Q$  of length  $n$  with  $r$  unmatched positions and

$$R = \bigcup_{\varepsilon \in \{0,1\}^r} \mathcal{A}_\varepsilon(P) \times \mathcal{A}_\varepsilon(Q).$$

Write the unmatched positions as

$$a_1 < \cdots < a_r \quad \text{in } P, \quad b_1 < \cdots < b_r \quad \text{in } Q.$$

Construct a matching  $M$  on  $[2n]$  as follows:

- include every edge of  $\mathcal{E}(P)$  in the left half;
- reflect every edge  $\{i, j\} \in \mathcal{E}(Q)$  to  $\{2n + 1 - j, 2n + 1 - i\}$  in the right half;
- for  $1 \leq i \leq r$ , add the cross-edge

$$\{a_i, 2n + 1 - b_i\}. \quad (31)$$

This is a perfect matching. It is noncrossing: internal edges are noncrossing; the right endpoints in (31) decrease as the left endpoints increase, so the cross-edges are nested; and an unmatched position of a Dyck prefix cannot lie strictly inside one of its matched edges, since it would still be on top of the stack when that edge closed.

For an internal edge of  $P$ , membership in a block  $\mathcal{A}_\varepsilon(P)$  means that the corresponding two first-half bits are opposite. The same is true for the reflected edges of  $Q$ , because complementing both endpoint bits preserves opposition. Finally, the common label  $\varepsilon$  gives

$$f_{a_i} = v(F)_{b_i} = 1 - f_{2n+1-b_i},$$

so the two bits on each cross-edge (31) are opposite. Hence every  $F \in \mathcal{F}$  belongs to  $\mathcal{F}_M$ . Both families have size  $2^n$ , so

$$\mathcal{F} = \mathcal{F}_M.$$

The matching is unique. Indeed, for every noncrossing perfect matching  $M$ ,

$$\{i, j\} \in M \iff w_i \neq w_j \text{ for every } w \in \mathcal{F}_M. \quad (32)$$

The forward implication is the definition. If  $i$  and  $j$  lie on different matching edges, orient those two edges independently so that  $w_i = w_j$ . Thus (32) recovers  $M$  from the family. Since the noncrossing perfect matchings of  $[2n]$  are counted by  $C_n$ , the number of extremal families is exactly  $C_n$ .  $\square$

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